

Do we price carbon?

In the discussion about climate change mitigation with possible economic gains, one approach countries can use is putting price on carbon through an Emissions Trading Scheme (ETS) or a Carbon tax. A crucial question arises, is carbon pricing even effective for emission reductions? On one hand, there are scholars demonstrating the positive environmental effects of carbon pricing, such as by Best et al. 2020, in the cross-country analysis including 31 countries, with growth rates of emissions in countries pricing carbon being around 2% lower than countries not doing so, with all else staying equal (Best et al. 2020, 91). Similar positive results of carbon taxes and carbon trading mechanisms have been discovered in other studies or reviews (Jia and Lin 2020; Metcalf 2021). With this being the case, authors also emphasize that for successful mitigation of climate change-related disasters, carbon pricing is only one method that should be present together with other policies (Fay et al. 2015, 2; Metcalf 2021). These additional requirements include additional investment into green technology innovation (Tinnereim and Mehling 2018), clean energy research (Metcalf 2021, 260) or establishing labels and performance standards (Fay et al. 2015, 20). Another argument in favour of carbon pricing as a good environmental tool is based on international cooperation, if done correctly (Roberts and Parks 2009). This is supported by the fact that further offsets in already largely decarbonized countries are much costlier in comparison to investment in developing countries, where the same decarbonization effect can be achieved much cheaper (Fursman et al. 2025, 49). Furthermore, this flow of finance can also be obtained from private actors, which are often able to amass far greater capital in comparison to bilateral or multilateral state-level cooperation (Kerr and Hu 2025).

On the other side of this argument are the scholars pointing at inherent issues of the carbon pricing mechanism. First line of arguments against carbon pricing is the essential incompatibility of our environmental goals and our gain-oriented monetary system (Leonardi 2017, 81). The second critique points out the non-urgent nature of carbon pricing mechanisms, emphasizing the necessity of quick action against climate change, with carbon pricing not only being ineffective but also counterproductive, as it creates the illusion of the climate being adequately addressed (Ball 2018, 146). Another critique falls on the unauthentic carbon price present across all the carbon markets due to the advertised price (supposed to advocate for legitimacy of the trading) being often disturbingly higher than the average price of the credits sold in the trading schemes (Finch and van den Bergh 2022). Another critique comes from Finon 2019, who suggests that carbon pricing specifically for developing economies is not adequate, as more political economy constraints are present in comparison to developed economies. Either way, carbon pricing measures have become key policy for many countries and are being promoted by the largest multilateral organizations, such as the United Nations 2022, World

Bank 2015 or Asian Development Bank 2024. As this is the case, it is crucial to examine what are the economic effects of carbon pricing measures.

Certain studies show potential GDP decreases in the case of implementation of carbon taxes without additional policies aiming to neutralize these effects (Calderón et al., 2016; United Nations 2022, 30). Due to this fact, it is, similarly to the environmental effects, crucial to implement carbon pricing measures in smart ways and accompany them by additional measures. By doing this, the negative economic effects are either annulled or even shifted into positive economic benefits. Perhaps the most crucial factor for the economic outcome of carbon pricing is smart usage of the new carbon-related revenue – revenue recycling (United Nations 2022). Multiple ways of using this new revenue have been tested and suggested, including broad reinvestment into all sectors (Ojha, Pohit, and Ghosh 2020), targeted cash transfers to households (Andriansyah and Hong 2022; Shapiro Finkelstein and Metcalf 2021) or targeted investment into infrastructure (Goulder 1995, 176). Another positive influence that carbon pricing may have is triggering innovation through achieving the “Porter effect,” as companies are incentivized to look for and invest in new, climate-friendly solutions (Dong et al. 2019).

How to price carbon?

Given the fragmented nature of the debate above, it is logical that the discussion regarding different carbon-pricing methods will be similarly polarized. ETSs have been criticized for multiple reasons. Besides the fact that an ETS only punishes emissions above a certain, often lenient, threshold, they have been critiqued for their, by other scholars praised, ability to generate mobilization of finance for developing economies as there is a danger of triggering a so-called “climate finance curse,” being compared to the resource curse (Jakob and Steckel 2014). Another argument against is the clear imbalance between carbon exports of developing countries and the revenue they receive for their emissions reductions, showcased by Pierucci et al. 2025 in the case of Brazil.

On the other hand, carbon tax has been praised for multiple things. Firstly, better results have been shown in the emission reduction capability when compared to ETS (Jia and Lin 2020). Secondly, they pose a better ability to recycle the newly obtained revenue (Advani et al. 2021; Ohja et al. 2020; Sunanda et al. 2025, 12). The third discussed advantage is the ease of implementation of the carbon tax, as it can be easily built upon existing energy and fuel taxes in developing countries (van den Bergh and Drews 2026). Furthermore, Metcalf 2021 praises its implementation, administration and compliance costs being significantly lower than in other measures. Lastly, scholars prefer the carbon tax due to its strong incentive for investment into low-carbon technologies, which is not necessarily present in the ETS with most projects focusing reducing deforestation and forest degradation (Advani et al. 2021). Despite the preference for carbon taxes, combination of the tax with an ETS may lead to the best results, as shown by Lin and Jia 2020.

Carbon pricing in Indonesia

As in other regions, carbon pricing is also the recommended measure capable of providing significant environmental and economic benefits, as it acts as immediate mitigation and provides a new revenue stream for green investments (Andriansyah and Hong 2022). Other studies argue for carbon pricing measures because of benefits for the environment (Chang and Phoumin 2022; Hasegawa et al. 2016; Kamandika and Dhakal 2023; Sujarwo 2016; Tan et al. 2022), the energy network (Chang and Phoumin 2022; Dutu 2016; Edianto et al. 2022),

renewable technologies (Chang and Phoumin 2022; Ordonez et al. 2022) or fiscal benefits through smart revenue reinvestment (Ordonez et al. 2022)

Surprisingly, more studies focus on the potential implementation of a carbon tax in Indonesia, although an ETS system has already been established and is expected to become fully functional this year. This may be related to the fact that an ETS has been discussed as less suitable for implementation in Indonesia due to unfavourable market conditions and political challenges, primarily in the energy and transportation sectors (Yusuf and Resosudarmo 2015; Kaneko and Kawanishi 2016). As this is the case, the main discussion about Indonesian carbon pricing revolves around establishing a carbon tax. One such option would be establishing a uniform carbon tax across ASEAN, which showed potential economic and environmental gains for Indonesia due to its heavy energy subsidies (Nurdianto and Resosudarmo 2016). Assessing carbon tax only in Indonesia, Yusuf and Resosudarmo 2015 suggest that in contrast to most industrialized country studies, carbon tax in Indonesia is not necessarily regressive as lower income households are less sensitive to price of energy-related commodities. More precisely, regressivity is projected to be present in urban areas but in rural ones, the effect is progressive at lower income levels and regressive at higher ones (Rahma et al. 2025). This theory is further supported by a case study of African countries, where higher-income households are, similarly to Indonesia, more likely to use fossil fuels and electricity, making the tax progressive at lower income rural areas (Advani et al. 2021).

Besides microeconomic effects on households and individuals, the macroeconomic picture for Indonesian implementation of carbon tax has also been debated. Hartono et al. 2023 argue that carbon tax revenue recycling combined with household transfer and investment resulted in higher GDP and employment. The relative progressivity of carbon tax in Indonesia together with potential GDP and employment gains suggest that it might be an adequate way of achieving decarbonisation while also achieving economic benefits. That said, the 2021 proposed Indonesian carbon tax is only around 2 USD per ton of CO₂, which is significantly below the international recommendations for developing countries of around 30-100 USD (Parry et al. 2021). Despite this being the case, there is still big potential for the carbon tax to develop into a successful tool for reducing emissions (Sunanda et al. 2025, 11). As showcased multiple times, this instrument is the preferred way of pricing carbon in the international and Indonesian context. To better understand the possibilities of this policy, the following section will provide an analysis of the proposed carbon tax with recommended adjustments to ensure the best environmental and economic benefits.

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